

Semester: VII						
DEEP LEARNING (GROUP-F: Professional Core Elective) (Theory)						
Course Code	:	CS374TFA		CIE	:	100 Marks
Credits: L:T:P	:	3:0:0		SEE	:	100 Marks
Total Hours	:	45L		SEE Duration	:	03 Hours

Unit-I	10Hrs
Neural Networks: What is a neural network, Models of a Neuron, Activation functions, Network Architectures, Knowledge representation, Learning Process. Deep Feedforward Networks: Multilayer Perceptron, Example: Learning XOR, Gradient-Based Learning, Hidden Units, Architecture Design, Back-Propagation Algorithm.	
Unit – II	9Hrs
Convolutional Networks: Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior, Variants of the basic convolution function, Structured Outputs, Data types, Efficient Convolution Algorithms, Random or Unsupervised features, The Neuroscientific basis for convolutional networks.	
Unit –III	10Hrs
Sequence Modeling: Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks, Echo State Networks, The Long Short-Term Memory and Other Gated RNNs.	
Unit –IV	9Hrs
Autoencoders: Undercomplete Autoencoders, Regularized Autoencoders, representational Power, Layer Size and Depth, Stochastic Encoders and Decoders, Denoising Autoencoders, Contractive Auto encoders, Applications of Autoencoders	
Unit –V	7Hrs
Pretrained models: Lenet, AlexNet, VGGNet, Densenet, Resnet, Improving Deep Neural Networks-Hyperparameter Tuning, Regularization and Optimization. Data Augmentation techniques. Other Architectures: Generative Adversarial Networks, Reinforcement Learning	

Course Outcomes: After completing the course, the students will be able to	
CO1:	Explore key theoretical concepts like the Universal Approximation Theorem, vanishing / exploding gradients, and optimization methods.
CO2:	Analyse the fundamental concepts of Deep Learning, and its various architecture learning models, including Neural Networks, backpropagation, gradient descent, and different Network Architectures (feedforward, convolutional, recurrent) Learning tasks for various applications.
CO3:	Apply the Deep learning model approaches to know the strengths and weaknesses of the architecture by empirical results. Apply appropriate concepts like Recurrent, Recursive Nets and Auto-encoder models to specific real time projects and analyse the Optimization techniques.
CO4:	Designing and implement a Deep Learning model as part of an experiential learning initiative in teams to solve societal and environmental problems. Ability to fine tune the model parameters to improve performance, explore and understand the ethical implications and societal impact of deploying deep learning systems in real-world scenarios.

Reference Books	
1	Deep Learning (Adaptive Computation and Machine Learning Series), Ian Good Fellow, Yoshua Bengio and Aaron Courville, MIT Press (3 January 2017), ISBN-13: 978-0262035613.
2	Neural Networks and Learning Machines, Simon S. Haykin, 3rd Edition 2010, PHI Learning, ISBN-9789332586253, 933258625X.
3	Introduction to Artificial Neural Networks, Gunjan Goswami, S.K. Kataria & Sons; 2012 Edition, ISBN-13: 978-9350142967.
4	Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms, Nikhil Buduma, by O'Reilly Publications, 2016 Edition, ISBN-13: 978-1491925614.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). Two tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) ADDING UPTO 40 MARKS.	40
MAXIMUM MARKS FOR THE CIE (THEORY)		100

RUBRIC FOR THE SEMESTER END EXAMINATION (THEORY)		
Q. NO.	CONTENTS	MARKS
PART A		
1	Objective type of questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100